

35) What can you say about $\text{Nul } B$ when B is a 5×4 matrix with linearly independent columns?

Solution: By defn, $\text{Nul } B$ represents the set of all ~~linear combinations of the columns of B~~ solutions x of the matrix equation $Bx=0$. Since the columns of B are linearly independent, $\therefore$ The matrix eqn. $Bx=0$ has only the trivial solution. $\therefore \text{Nul } B = \{0\}$ $\square$

36) What can you say about the shape of an $m \times n$ matrix A when the columns of A form a basis for $\mathbb{R}^m$?

Solution: By defn. of basis, the columns of A are linearly independent and they span $\mathbb{R}^m$. By Theorem 4 from Chapter 1, the ~~column~~ matrix A has a pivot position in every row. By defn, the pivot position of a ~~row~~ matrix corresponds to the leading 1 of a row in rref. By the Row Reduction Algorithm and Theorem 1 from Chapter 1, A can be converted to a unique rref. By Condition 2 from rref, for each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it. By Condition 4 from rref, each leading entry must be 1. $\therefore$ There must exist at least m columns of A each having 1 pivot.

So, $m \leq n$ $\square$